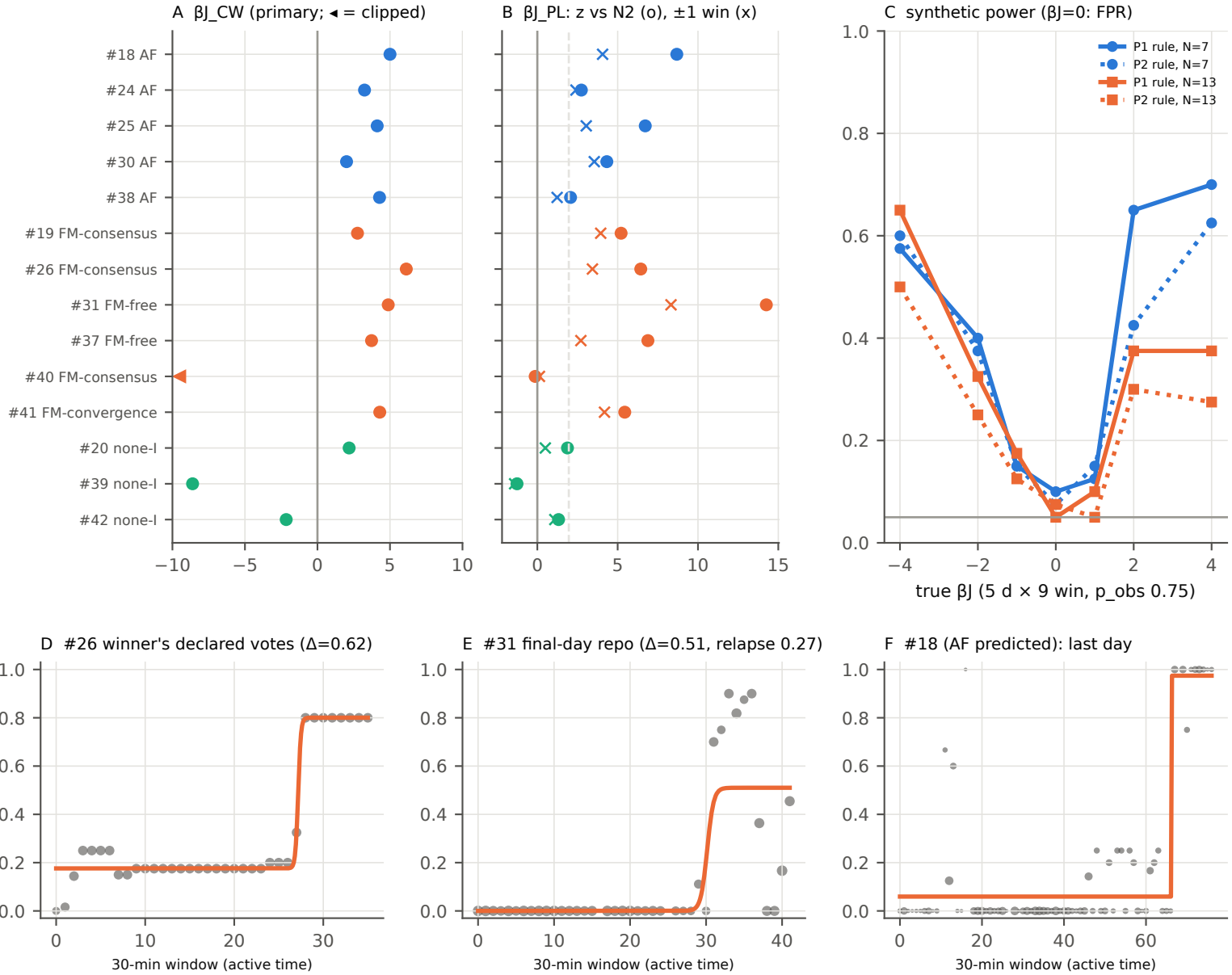


H11 · Division of labor vs herding as the sign of a Potts coupling

Exploratory round 1 · non-holdout goal periods · 2026-10-03

Headline: the sign of the uniform-coupling (mean-field) Potts coupling does NOT follow the goal's mode. Project choice is ferromagnetic ($\beta_{J_CW} > 0$) in 11 of 14 tested weeks — including all five shared-objective 'divisible' weeks predicted to be antiferromagnetic ($\beta_{J_CW} + 2.0 \dots + 5.0$; coupling beyond agent fields $z_{N2} + 2.1 \dots + 8.7$). Spread ('division of labor') appears only as static specialization in the three weeks where the goal gave each agent its own artifact (#39 worlds, #42 channels, #40 worlds + hub). Sign test of the mode rule: 5/11 correct ($p = 0.73$); AF<FM contrast $p = 0.47$. Consensus: the #26 runoff is an abrupt, persistent jump in declared votes ($\Delta = 0.62$ within one 30-min window), but its mean-field coupling sits at the first-order threshold with a wide CI ($\beta_J = 2.8 [0.65, 4.65]$ vs $\beta_{J_S} = 2.75$); #31 shows herding waves, not one consensus.



Prediction

- P1 sign of β_{J_CW} by mode (card level)
- P2 coupling beyond agent fields by mode
- P3 first-order consensus (#19, #26, #31, #40)
- P4 action class does not follow mode
- P5 held-out gain where $|z_{N2}| \geq 2$
- P6 plmDCA agrees with β_{J_PL}
- P7 $|\beta_J| < 2$ everywhere (prior)
- HH22 P-G26a / b / c

Outcome

- MIXED by rule (4 supported, 2 failed, 1 n/a of 7); AF half fails 0/5
- FM: 5/6 $z \geq 2$; AF: 5/5 $z \geq +2$ (opposite) → failed
- 0/4 full passes; #26 votes: jump yes, MF at threshold
- supported (contrast $p = 1.0$; no sig. negatives)
- 8/10 periods
- agrees in sign (null absorbs no drift)
- failed for β_{J_CW} (12/14); within-day excess 0.4–3.4
- uninformative pass / supported / inconclusive

Data: strict artifact mentions (H07 tables), 30-min windows, room blocks; 14 non-holdout periods tested, 3 n/a. Estimators: exact finite-N Curie-Weiss Potts MLE (generating functions), agent-field pseudo-likelihood vs circular-shift null. Caveats: common within-day drive and coupling are not separable; specialization reads as AF in β_{J_CW} ; per-period power ≈ 0.35 at $|\beta_J| = 2$. Holdout untouched; confirmatory targets #22, #28, #45 (script written, not run). Code: hypotheses/H11-potts-labor-vs-herding/analysis/.